

GROZNY, Russian Federation

WMO: 372350

Lat: 43.250N

Lon: 45.717E

Elev: 845

StdP: 14.25

Time Zone: 3.00 (E03)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.7	10.9	1.6	6.1	6.6	6.5	7.9	11.9	27.6	36.7	22.6	29.3	3.5	290	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	91.9	70.2	88.9	69.2	86.2	68.6	73.3	86.2	71.6	83.8	70.3	81.9	7.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	111.2	78.6	67.9	105.8	76.8	66.6	100.9	75.7	37.5	86.9	36.0	83.6	34.8	82.3	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.1	18.6	16.0	DB	0.5	96.7	8.9	3.8	-5.9	99.4	-11.1	101.6	-16.1	103.7	-22.6	106.5	
			WB	0.0	77.9	8.8	3.2	-6.3	80.3	-11.5	82.1	-16.4	84.0	-22.8	86.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	29.4	30.2	40.5	51.9	61.7	70.9	75.4	73.7	64.9	51.8	39.4	31.7	(6)
	DBStd	17.96	7.33	9.32	6.59	6.40	6.00	5.45	4.87	4.66	6.33	6.87	8.57	7.73	(7)
	HDD50	2503	637	556	302	54	3	0	0	0	1	61	323	566	(8)
	HDD65	5667	1102	975	761	392	136	11	1	2	78	409	769	1031	(9)
	CDD50	3202	0	1	7	112	365	627	788	734	447	117	4	0	(10)
	CDD65	894	0	0	0	1	33	188	325	271	75	1	0	0	(11)
	CDH74	7705	0	0	0	23	264	1569	2963	2322	547	17	0	0	(12)
	CDH80	2779	0	0	0	2	36	551	1202	869	118	1	0	0	(13)
	WSAvg	5.7	5.0	4.9	6.3	7.4	6.9	5.9	6.1	5.5	5.3	5.5	5.4	4.5	(14)
(15) Precipitation	PrecAvg	22.3	1.7	1.5	1.7	1.7	2.4	2.8	2.1	1.8	1.9	1.7	1.6	1.6	(15)
	PrecMax	30.0	3.4	3.4	2.9	3.0	4.0	5.0	3.8	3.4	6.1	3.6	3.7	3.8	(16)
	PrecMin	17.4	0.2	0.7	0.9	0.6	0.6	0.9	0.7	0.4	0.2	0.3	0.1	0.4	(17)
	PrecStd	3.4	0.7	0.6	0.6	0.6	0.9	1.1	0.8	0.8	1.3	0.8	0.9	0.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.8	58.0	68.0	77.1	83.7	93.9	97.2	95.2	87.6	76.4	61.6	51.4	(19)
		MCWB	42.4	47.4	53.0	59.4	65.8	70.5	72.8	70.6	67.8	60.6	52.4	45.1	(20)
	2%	DB	43.0	47.8	60.9	72.5	80.0	89.3	92.8	90.8	83.3	70.8	56.1	46.7	(21)
		MCWB	38.3	42.1	48.7	57.8	64.4	68.8	71.8	69.6	66.7	58.7	50.1	42.1	(22)
	5%	DB	39.6	43.1	55.7	68.8	76.9	86.1	89.4	87.8	80.0	66.8	52.2	43.3	(23)
		MCWB	36.9	39.0	46.8	55.6	63.1	68.2	69.4	69.1	65.3	57.0	47.7	40.5	(24)
	10%	DB	37.2	39.7	51.1	65.2	73.8	82.6	86.6	84.8	76.8	62.8	49.3	40.6	(25)
		MCWB	35.2	37.3	44.6	53.9	62.0	66.9	69.1	68.0	64.5	55.4	46.4	38.6	(26)
	0.4%	WB	43.1	48.1	53.1	61.1	68.6	73.4	76.4	74.8	71.0	62.8	54.2	46.2	(27)
		MCDB	48.2	55.5	67.3	73.7	80.0	87.6	91.7	86.5	81.8	70.8	57.6	50.8	(28)
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.3	42.5	49.6	59.0	66.5	70.9	74.3	72.5	69.0	59.9	50.7	43.1	(29)
		MCDB	42.0	46.6	58.3	71.2	76.5	85.1	88.3	85.1	78.8	68.7	55.4	45.9	(30)
	5%	WB	37.2	39.5	47.4	56.8	64.5	69.4	72.3	71.0	67.3	57.9	48.4	41.0	(31)
		MCDB	39.0	42.4	54.7	66.6	74.1	82.5	84.2	83.1	76.6	64.4	51.1	42.6	(32)
	10%	WB	35.5	37.6	45.0	54.7	62.8	68.1	70.9	69.7	65.6	56.2	46.5	38.7	(33)
		MCDB	37.0	39.5	50.6	63.3	71.7	79.8	82.0	80.9	74.2	61.9	49.2	40.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	11.1	13.9	19.1	18.6	18.9	19.1	19.6	18.7	17.2	11.0	10.1	(35)
		MCDBR	14.6	16.8	25.5	27.1	23.6	24.3	23.7	24.6	24.4	25.2	17.7	14.4	(36)
		MCWBR	10.4	10.9	14.3	13.5	10.3	8.9	7.7	7.8	10.0	13.2	11.1	10.0	(37)
	5% WB	MCDBR	12.3	15.3	23.3	24.1	21.1	21.7	20.7	21.3	20.6	21.9	15.6	12.5	(38)
		MCWBR	9.3	10.2	13.4	12.6	10.2	9.0	8.4	8.5	9.6	12.0	10.4	9.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.331	0.365	0.412	0.466	0.441	0.447	0.482	0.479	0.432	0.385	0.348	0.333	(40)
	taud		2.338	2.232	2.142	2.035	2.190	2.225	2.146	2.147	2.248	2.336	2.372	2.336	(41)
	Ebn at Noon		249	258	259	254	265	263	252	248	251	249	241	236	(42)
	Edh at Noon		26	35	43	51	45	44	47	46	38	31	25	24	(43)
(44) All-Sky Solar Radiation	RadAvg		452	702	999	1406	1705	1939	1914	1748	1318	825	525	393	(44)
	RadStd		67	125	121	164	177	123	149	141	133	109	75	56	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page